

Near-Gaze Visualisations of Empathic Communication Cues in Mixed Reality Collaboration

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ABSTRACT

In this poster, we present a live 360° panoramic-video based empathic Mixed Reality (MR) collaboration system that shares various Near-Gaze non-verbal communication cues including gaze, hand pointing, gesturing, and heart rate visualisations in real-time. The preliminary results indicate that the interface with the partner's communication cues visualised close to the gaze point allows users to focus without dividing attention to the collaborator's physical body movements yet still effectively communicate. Shared gaze visualisations coupled with deictic languages are primarily used to affirm joint attention and mutual understanding, while hand pointing and gesturing are used as secondary. Our approach provides a new way to help enable effective remote collaboration through varied empathic communication visualisations and modalities which covers different task properties and spatial setups.

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1 INTRODUCTION

Face to Face collaboration allows pairs of people to easily access both implicit and explicit communication cues including gaze, speech, and gestures. These dynamic interactions are challenging when collaborating remotely even via video conferencing. To overcome such limitations, researchers have tried using Mixed Reality (MR) to recreate a virtual shared task environment through varied modalities, incorporating wearable sensors to visualise real-time biofeedback (gaze, heart rate) together with physical hand and body movements, resulting in novel augmentation of human empathy. Previous studies have suggested that visualising physiological cues

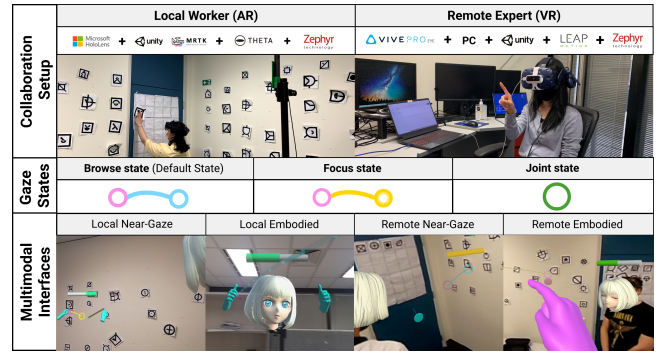


Figure 1: System Overview

(e.g. heart rate, and respiration rate) in multimodal remote collaboration improves co-presence and empathic experience to enhance task performance. However, to the best of our knowledge, no research has investigated how real-time visualisations of non-verbal communication cues, such as gesture and biofeedback, should be presented with gaze behaviours [Jing et al. 2021b] in an MR remote collaboration context. In a physical task space, view positions and the distance to the target objects may skew the interpretation and mutual understanding of hand pointing and gesture communication. Collaborators facing different directions may also need to split attention between the user's own task view and the partner's location. As a result, presenting the partner's task-related visual and biofeedback cues next to their current gaze point on the surface of interest, compared to the conventional method of displaying them at their original location, may stimulate different empathic collaboration experiences, especially between pairs who work remotely on a task requiring a high level of mental effort and least divided attention. In this poster, we present a MR collaboration system that visualises various empathic communication cues at the user's gaze point (Near-Gaze). These include gaze, pointing, gestures, and physiological visualisations. Our prototype allows users to explore how each modality influences each other and what level of task setup affects empathic experience during MR remote collaboration.

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2 PROTOTYPE DESIGN

Our prototype (Figure 1) offers gaze-based multimodal interactions combining gaze behaviours, hand pointing, gesturing, and heart rate visualisations through a live 360° panoramic video-based collaborative MR environment. Users are able to freely exchange bi-directional real-time shared gaze cues and other communication modalities between a local worker and a remote expert, while being able to constantly monitor each others' heart rate changes.

360° Mixed Reality View. Figure 1: Collaboration Setup, the local worker in the immediate task space wears an Augmented Reality (AR) head-mounted display (HMD). S/he can take physical actions on the surrounding objects and a live view will be streamed to the remote expert via a 360° panoramic camera placed at the centre of the task space. The remote expert wearing a Virtual Reality (VR) HMD can also exchange the same communication cues with the local worker. Both views are omni-directional and independent from each other.

Shared Gaze Representation. Three gaze behaviour (Figure 1: Gaze States) are visualised: *Browse state (default)*: active eye movements. *Focus state*: gaze dwells for 500ms or more. *Joint state*: collaborators' gazes overlap for over 200ms. For the Browse and Focus states, a Joint Gaze Indicator is made available to link between both user's and collaborator's gaze points for easy identification [Jing et al. 2021a].

Hand Pointing and Gesturing. As shown in Figure 1: Multimodal Interfaces, both local and remote users can point with their right index fingers to an object of interest in the task space. A virtual ray is shown casting from the tip of the index finger pointing towards the target surface. The remote expert's hand gestures are tracked and visualised as a virtual hand shown in the local worker's HMD. Since the local worker's gestures can be directly captured and viewed through the live video, a virtual hand of the local's is considered unnecessary from the remote expert's HMD, and only the pointing ray is visualised.

Heart Rate Visualisation. The real-time heart rates (HR) of both collaborators are represented as an animated 3D bar, updated every second. The design currently includes HR in four states: *Normal state (default)*: in green colour for HR between 60 100bpm where the bar increments by 1% every unit. *Warning state (80% of the max HR)*: in yellow colour for 100bpm < HR < 140bpm or 40bpm < HR < 60bpm. *Fatal state (90% of the max HR)*: in red colour for HR > 140bpm or HR < 40bpm. *Error stage*: in grey colour when HR sensor fails to track properly.

Near-Gaze Visualisation. In the conventional method, a user's partner's eyes, hand, and physiological cues are visualised based on their body at their estimated location (Figure 1: *Embodied*). In contrast, visualisation cues are placed close to the current gaze focus of the user's for easy identification in the proposed *Near-Gaze* method, therefore minimising the split of attention between the current task space and the partner's location.

Implementation. The system consists of two parts: *Local* - a HoloLens2 see-through HMD with built-in eye and hand tracking worn by the local worker, a Ricoh Theta V 360° panorama live

camera (1080p) to capture the task space, and a Vuforia tracking marker attached to the 360° camera to align the AR-VR coordinate systems. *Remote* - a Vive Pro Eye HMD worn by the remote expert, which a Leap Motion sensor attached to track hand gestures, and a PC (Intel Core i7-10870H 2.20GHz CPU, 16GB DDR4 RAM, NVIDIA GeForce RTX 3060 GPU, Windows 10 OS) running the Unity game engine (v 2019.3.15). The transmitted 360° scene of the task environment is mapped onto a sphere which the remote user views from the centre. Each collaborator wears a Zephyr BioHarness3 sensor to track ECG data during the task. We used the MRTK 2.4.0, SRanipal SDK and Ultraleap Gemini libraries to visualise gaze and gesture cues. All visualised information was exchanged via WiFi and displayed synchronously between users.

3 EXPLORATORY STUDY

We conducted expert evaluations with eight participants, known each other as "work colleagues", paired into four groups. The study explored a 2x2 experimental design with two levels of interfaces (*Embodied*, *Near-Gaze*) and two levels of roles (local novice worker, remote expert collaborator), with an asymmetrical remote collaboration on a puzzle-solving task. We used surveys, interviews, and observations to understand user experience. The results indicated that participants preferred shared gaze coupled with deixis as a primary pointing method, with hand pointing and gesturing as secondary confirmation. Collaborators found *Near-Gaze* UI more suitable to avoid attention division. Physiological visualisations might work better in tasks requiring a remote expert to adjust communication style based on local novice user's cognitive ability.

4 CONCLUSIONS AND FUTURE WORK

We present a live 360° panoramic-video based Mixed Reality collaboration system sharing *Near-Gaze* non-verbal communication cues and biofeedback such as gaze, hand pointing, gesturing, and heart rate between remote collaborators in real-time. Preliminary user study results suggest that the *Near-Gaze* visualisations allow users to focus on target object without dividing attention from the collaborator's physical body and movements, while the *Embodied* visualisations provide natural full body continuous interaction. Additionally, in the free-formed task space, participants primarily relied on shared gaze coupled with deictic languages to affirm mutual understanding, followed by hand pointing and gesturing as a secondary method. Although not actively used by our participants, physiological visualisations also exert future investigation within tasks that require more empathic communication.

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